A partial function $f : \mathbb{N}^n \rightharpoonup \mathbb{N}$ is **λ-definable** *iff* there exists a closed λ-term M with the following property:

$$f(x_1, \ldots, x_n) = y \quad \textit{iff} \quad M\ \underline{x_1}\ \underline{x_2} \ldots \underline{x_n} =_\beta \underline{y}$$

and

$$f(x_1, \ldots, x_n) \uparrow \quad \textit{iff} \quad M\ \underline{x_1}\ \underline{x_2} \ldots \underline{x_n} \text{ has no normal form}$$

where $\underline{n}$ denotes the encoding of the natural number n in the λ-calculus.